CMTFNet: CNN and Multiscale Transformer Fusion Network for Remote-Sensing Image Semantic Segmentation

Honglin Wu, Peng Huang, Min Zhang, Wenlong Tang, and Xinyu Yu

***Abstract*— Convolutional neural networks (CNNs) are powerful in extracting local information but lack the ability to model long-range dependencies. In contrast, the transformer relies on multihead self-attention mechanisms to effectively extract the global contextual information and thus model long-range dependencies. In this article, we propose a novel encoder–decoder structured semantic segmentation network, named CNN and multiscale transformer fusion network (CMTFNet), to extract and fuse local information and multiscale global contextual information of high-resolution remote-sensing images. Specifically, to further process the output features from the CNN encoder, we build a transformer decoder based on the multiscale multihead self-attention (M2SA) module for extracting rich multiscale global contextual information and channel information. Additionally, the transformer block introduces an efficient feed-forward network (E-FFN) to enhance the information interaction between different channels of the feature. Finally, the multiscale attention fusion (MAF) module fully fuses the feature information from different levels. We have conducted extensive comparison experiments and ablation experiments on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam datasets. The extensive experimental results demonstrate that our proposed CMTFNet can obtain superior performance compared to the currently popular methods. The codes will be available at https://github.com/DrWuHonglin/CMTFNet.**

***Index Terms*— Global contextual information, multiscale transformer, remote-sensing image, semantic segmentation.**

I. Introduction

SEMANTIC segmentation of high-resolution remote-sensing images is a hot topic in the field of remote-sensing image processing. It provides support for a variety of remote-sensing image application scenarios, including urban planning [1], [2], [3], land cover classification [4], [5], [6], land planning [7], and road extraction [8]. Semantic segmentation is the pixel-level classification of the entire image and labels the category to which each pixel in the image belongs with different markers. However, high-resolution remote-sensing images have complex backgrounds, rich feature details, and the characteristics of large intraclass variance and small interclass variance. Traditional semantic segmentation methods such as support vector machines [9], random forest [10], and conditional random field [11] have been difficult to meet the current demand for high-resolution remote-sensing image applications. Therefore, semantic segmentation of high-resolution remote-sensing images remains a challenging task.

With the rapid development of deep-learning techniques, many researchers have proposed convolutional neural network (CNN)-based approaches to solve the problems in the semantic segmentation task of high-resolution remote-sensing images. In particular, Long et al. [12] proposed the fully convolutional network (FCN), which accomplishes true pixel-level segmentation and significantly improves segmentation accuracy. Although FCN obtains favorable results, its overly simple upsampling leads to less refined segmentation results. Thereafter, Ronneberger et al. [13] proposed UNet, an encoder–decoder network with a symmetric structure. The segmentation accuracy was further improved by compensating the feature information lost by downsampling through skip connection. However, UNet simply splices the low-level features generated by the encoder with the high-level features generated by the decoder without further refining the features, thus limiting the network's ability to distinguish between low-level and high-level features.

Recently, some related researchers have started to notice the importance of multiscale features and attention mechanisms in semantic segmentation. Chen et al. [14], [15], [16], [17] proposed the DeepLab series of networks, which increases the receptive field without reducing the image size by applying the atrous convolution and proposed the atrous space pyramid pooling (ASPP) module for multiscale feature extraction. They also introduce the depthwise separable convolution and Xception modules to reduce the computational and parametric quantities with segmentation performance improvement. Zhao et al. [18] proposed the pyramid scene parsing network (PSPNet), which uses global average pooling to construct a pyramid pooling module (PPM) to capture features at different scales and fuse them, thereby making full use of information at different scales to improve segmentation accuracy. Liu et al. [19] proposed with scale-feature attention module (SFAM) to enhance deep and

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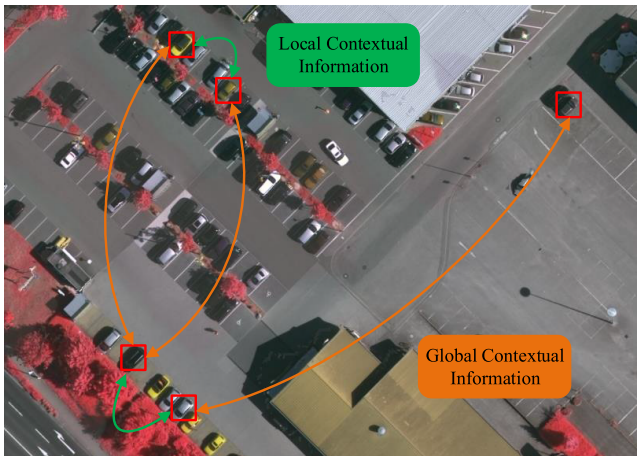

Fig. 1. Illustration of the local and global contextual information.

shallow layer features and learn representative features using a multiscale structure with a scale-layer attention module (SLAM), which helps the network to distinguish objects for semantic segmentation. Li et al. [20] proposed a semantic segmentation network with spatial and channel attention mechanism (SCAttNet), which combines SCAttNets to establish feature associations and improve the accuracy of semantic segmentation. Niu et al. [21] proposed a new attention-based framework called hybrid multiple attention network (HMANet) to improve segmentation performance by adaptively capturing global relevance from spatial, channel, and category perspectives in an efficient manner. Therefore, utilizing multiscale features and attention mechanisms to enhance network performance in semantic segmentation tasks is a common and effective approach.

Despite these advantages, CNNs usually rely on convolutional operations for feature extraction. Due to the limitations of convolution, each convolutional kernel focuses mainly on local information within its perceptual domain, making it difficult to model long-range dependencies. In addition to local context information, global context information is crucial for semantic segmentation, as shown in Fig. 1. With the rapid development of transformers in the field of natural language processing (NLP), some researchers have tried to solve this problem by using transformers in semantic segmentation tasks [22], [23], [24], [25]. Transformer models long-range dependencies by using multihead self-attention mechanisms to capture global contextual information about the image. Although the transformer has powerful global contextual information extraction capability, it also requires a large amount of memory and computational resources. The computational complexity of multihead self-attention is $O(n^2)$ of the image size, which makes it difficult for the transformer to handle high-resolution remote-sensing images. To reduce the computational complexity of transformer-based structures, Liu et al. [26] proposed the hierarchical transformer structure Swin transformer. This network restricts the self-attention computation to nonoverlapping local windows and establishes cross-window connections by shifting windows, significantly improving the self-attention computation efficiency. Wang et al. [27] proposed the pyramid vision transformer (PVT) to reduce the computational overhead by using a spatial-reduction attention mechanism (SRA). These methods reduce computational complexity by artificially dividing windows or using individual pooling for downsampling, thereby reducing the length of the sequence, but also weakening feature extraction capability and ignoring the channel information and multiscale information of the features.

Inspired by the above literature, we propose a CNN and multiscale transformer fusion network (CMTFNet) based on the encoder–decoder structure for remote-sensing image semantic segmentation. On the encoding side, we exploit CNN-based ResNet [28] to extract local information. On the decoding side, we propose a new self-attention module called multiscale multihead self-attention (M2SA). In contrast to the traditional multihead self-attention module, we embed multiscale and channel information into the self-attention to efficiently learn multiscale global information. Based on M2SA, a multiscale transformer decoder is developed to capture rich contextual information at a low computational cost. Furthermore, we propose an efficient feed-forward network (E-FFN) to replace the feed-forward network (FFN) in traditional transformers to enhance information interaction between channels. Finally, we design a multiscale attention fusion (MAF) module to fully fuse the feature information from different levels to achieve adaptive fusion of deep and shallow features. The main contributions can be summarized as follows.

1) We propose a novel semantic segmentation network CMTFNet. CMTFNet uses the CNN and multiscale transformer as the encoder and the decoder to extract local–global contextual information and fulfill multiscale feature learning and fusion. Experiments on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam datasets show that the method achieves a superior semantic segmentation effect with low complexity.
2) By integrating multiscale information and channel information into self-attention mechanisms, we propose a low computational cost M2SA module for extracting rich multiscale global contextual information and channel information. Moreover, the information interaction between different channels is further enhanced by introducing the E-FFN module to achieve better segmentation performance.
3) We propose an MAF module for adaptively fusing deep and shallow features extracted by the encoder and the decoder to achieve efficient aggregation of features.

II. Related Work

A. CNN-Based Semantic Segmentation of Remote-Sensing Images

FCN [12] achieves end-to-end semantic segmentation by employing transpose convolution instead of the fully connected (FC) layer in traditional networks. The proposed FCN is a milestone in CNN-based semantic segmentation methods. Since then, CNN-based methods have dominated the field of remote-sensing image semantic segmentation [29], [30]. In this

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process, the encoder–decoder structure exhibits excellent segmentation performance. For example, Ronneberger et al. [13] proposed UNet. Badrinarayanan et al. [31] proposed SegNet. The encoder extracts the hierarchical features by gradually reducing the feature resolution, and the decoder gradually restores the feature resolution while fusing high-level semantic features and low-level spatial information. Subsequently, the encoder–decoder architecture became the dominant framework for semantic segmentation of remote-sensing images [32]. Although the CNN approach based on the encoder–decoder structure has achieved outstanding results. However, ground objects with varying scales and high similarity characteristics pose new challenges to remote-sensing image semantic segmentation. Therefore, identifying these complex targets requires finer spatial features and global contextual information.

To address this problem, existing methods apply multiscale and attention mechanisms to model long-range dependencies and feature refinement. Li et al. [33] proposed an attention aggregation feature pyramid network (A^2-FPN). A^2-FPN learns multiscale features through a feature pyramid network (FPN) and uses an attention aggregation module (AAM) to enhance multiscale feature learning. Bai et al. [34] proposed a multiscale attention (MSA) and nonlocal filter network (MsanlfNet). The MSA module enhances the fine-grained multiscale representation capability of the network, while the nonlocal filter (NLF) module captures global contextual information. Zhao et al. [35] proposed a semantic segmentation with an attention mechanism network (SSAtNet), which uses a pyramid attention pooling module to introduce the attention mechanisms into the multiscale module for adaptive feature refinement. Li et al. [36] designed a multistage attention reconstruction network (MAResU-Net), where long-range dependencies are established by adding a linear attention mechanism (LAM) to the skip connections of the original UNet. Li et al. [37] proposed a multiattention network (MANet) to extract contextual dependencies through multiple efficient attention modules. Although the above methods achieve outstanding results to a certain extent, these methods obtain global information through aggregating the local features extracted by CNN, instead of modeling global information directly. Therefore, the extracted global information may not be comprehensive.

B. Transformer-Based Semantic Segmentation of Remote-Sensing Images

In recent years, with the rise of transformers in computer vision tasks, an increasing number of researchers have begun to focus on its application in the field of semantic segmentation. By learning the relationship between sequence elements, the transformer can effectively capture global contextual information. Existing transformer-based methods for remote-sensing image semantic segmentation can be broadly classified into two categories. The first category is an encoder–decoder architecture consisting of pure transformers, such as Segmenter [38], SegFormer [39], and Swin-UNet [40]. The second category is a hybrid encoder–decoder architecture of the CNN and transformer. Most existing methods typically employ a hybrid design that aims to combine the merits of CNNs and transformers. Xu et al. [41] proposed a lightweight transformer-based network Efficient-T, which uses a hierarchical Swin transformer and MLP segmentation head to achieve a tradeoff between computational effort and accuracy. Wang et al. [42] proposed a bilateral awareness network (BANet) with a dependency path and texture path, where the dependency path captures long-range dependencies based on the transformer and the texture path extracts local texture features based on CNN. Zhang et al. [43] proposed a hybrid architecture where the encoder uses a Swin transformer to capture remote dependencies, while the decoder uses a CNN-based approach to better preserve local information in the image.

III. Methodology

In this section, we will introduce the architecture and the component modules of CMTFNet in detail. The general structure of the network is introduced in Section III-A, the M2SA module is introduced in Section III-B, the E-FFN module is introduced in Section III-C, and the MAF module is introduced in Section III-D.

A. Overall Structure

The overall network structure of the CMTFNet is shown in Fig. 2. It consists of two parts: the encoder and the decoder. For the encoder, we used ResNet50 as the backbone network for extracting the hierarchical features. ResNet50 consists of four stages, each ResBlock stage includes three layers, which are 1×1 convolution, 3×3 convolution, and 1×1 convolution. Each stage was downsampled by a scale factor of 2, resulting in a total of four feature maps of different resolution sizes: $H/4 \times W/4$, $H/8 \times W/8$, $H/16 \times W/16$, and $H/32 \times W/32$. For the decoder, the four output features of the encoder are first unified to 512 channel dimensions using 1×1 convolution, and fused with the corresponding feature maps of the decoder by the Fusion module. Specifically, the local semantic features generated by ResBlock and the global contextual features generated by the decoder are weighted and fused using a learnable parameter α. The weighted summation operation adaptively adjusts the size of the weights according to the contribution of these two features, which can be expressed as

$$\boldsymbol{F} = \alpha \cdot \mathbf{RF} + (1 - \alpha) \cdot \text{Upsample}(\mathbf{MTF}) \tag{1}$$

where **RF** denotes the output features of the ResBlock, **MTF** denotes the output features of the multiscale transformer block, Upsample$(\cdot)$ denotes $2 \times$ upsampling, and $\boldsymbol{F}$ denotes the fused features.

Then, the output of ResBlock4 is fed into a multiscale transformer block to extract multiscale global contextual information. It is first inputted into the M2SA module and then passed on to the normalization layer, whose output is residually connected with the original input. Here, we use the BatchNorm rather than the LayerNorm in the traditional transformer [18] for better adapting to the 2-D image structure.

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Its output is passed through the E-FFN module and residual connection to get the final output. The above process can be expressed as follows:

$$\begin{aligned} \boldsymbol{X}_{\text{att}} &= \boldsymbol{X} + \text{BN}(\text{M2SA}(\boldsymbol{X})) \\ \boldsymbol{X}_{\text{out}} &= \boldsymbol{X}_{\text{att}} + \text{E-FFN}(\boldsymbol{X}_{\text{att}}) \end{aligned} \tag{2}$$

where $\boldsymbol{X}$, $\boldsymbol{X}_{\text{att}}$, and $\boldsymbol{X}_{\text{out}}$ are the input of M2SA, the output of the first subblock, and the output of multiscale transformer block, respectively. The output of the last Fusion module is sent to Seghead for prediction, first into the MAF module for feature aggregation, and then through a 3×3 convolution and $4 \times$ upsampling to obtain the final predicted image.

B. M2SA Module

The global information of the image is important for semantic segmentation. To extract more powerful global information, we propose the M2SA module as shown in Fig. 3. It integrates multiscale mechanisms, channel attention mechanisms, and multihead self-attention mechanisms.

We apply two branches on the input feature $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$ to extract multiscale global contextual information and channel information, where C denotes the channel dimension, and H and W denote the size of the image. Specifically, at the first branch, the channel dimension C of the input feature $\boldsymbol{X}$ is first reduced by a factor of 4 through a 1×1 convolution and then fed into three parallel depthwise separable atrous convolutions to extract multiscale information, where the convolution kernel size is 3×3, and the atrous rate R_i ($i = \{1, 2, 3\}$) is $\{(1, 3, 5), (3, 5, 7), (5, 7, 9)\}$ corresponding to the M2SA modules in each of the three multiscale transformer blocks. Then, the original channel dimension is recovered by a 1×1 convolution, and the results of the three branches are fused by a summation operation. Finally, the output is obtained by a 1×1 convolution and residual connection. In summary, the above process can be expressed as

$$\begin{aligned} \boldsymbol{X}_i &= \text{dwConv}_{3\times3}^{R_i}(\text{Con}v_{1\times1}(\boldsymbol{X})) \\ \tilde{\boldsymbol{X}}_i &= \text{Con}v_{1\times1}(\boldsymbol{X}_i) \\ \boldsymbol{X}_{\text{out}} &= \text{Con}v_{1\times1}\left(\tilde{\boldsymbol{X}}_1 + \tilde{\boldsymbol{X}}_2 + \tilde{\boldsymbol{X}}_3\right) \otimes \boldsymbol{X} \end{aligned} \tag{3}$$

where $\otimes$ denotes element-wise multiplication, $\boldsymbol{X}_i$ ($i = 1, 2, 3$) denotes the output of three depthwise separable atrous convolutions, respectively, $\text{Con}v_{1\times1}(\cdot)$ denotes a 1×1 convolution, $\text{dwConv}_{3\times3}^{R_i}(\cdot)$ denotes a 3×3 depthwise separable atrous convolution with an atrous rate of R_i, $\tilde{\boldsymbol{X}}_i$ ($i = \{1, 2, 3\}$) denotes the output of the three 1×1 convolution branches, and $\boldsymbol{X}_{\text{out}}$ denotes the final output. Afterward, we feed the resulting features into an adaptive average pooling layer

$$\boldsymbol{P} = \text{AdaptivePool}(\boldsymbol{X}_{\text{out}}) \tag{4}$$

where, for simplicity, the flattening operation is omitted. In this way, the obtained feature $\boldsymbol{P} \in \mathbb{R}^{C\times P\times P}$ has a smaller resolution than the original input $\boldsymbol{X}$, where P is set to 16. In addition, $\boldsymbol{P}$ contains the rich multiscale contextual information of input $\boldsymbol{X}$ and can thus act as the alternative to input $\boldsymbol{X}$ when computing multihead self-attention.

Denote the query, key, and value tensor in multihead self-attention by $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$, respectively. In contrast to the traditional calculation method

$$(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \left(\boldsymbol{X}\boldsymbol{W}^q, \boldsymbol{X}\boldsymbol{W}^k, \boldsymbol{X}\boldsymbol{W}^v\right) \tag{5}$$

we use

$$(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \left(\boldsymbol{X}\boldsymbol{W}^q, \boldsymbol{P}\boldsymbol{W}^k, \boldsymbol{P}\boldsymbol{W}^v\right) \tag{6}$$

where $\boldsymbol{W}^q$, $\boldsymbol{W}^k$, and $\boldsymbol{W}^v$ represent the weight matrix of linear transformation used to generate query, key, and value tensors, respectively. Then, $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are fed into the multihead self-attention module to compute the self-attention feature, which is formulated as follows:

$$\text{Attention} = \text{Softmax}\left(\frac{\boldsymbol{Q} \times \boldsymbol{K}^T}{\sqrt{d_k}}\right) \times \boldsymbol{V} \tag{7}$$

where d_k is the channel dimension of $\boldsymbol{K}$, and division by $\sqrt{d_k}$ can be considered as an approximate normalization. The Softmax function is calculated for each row in the matrix. Equation (6) omits the concept of multiple heads [22], [44] for simplicity. Since $\boldsymbol{K}$ and $\boldsymbol{V}$ have a shorter length than $\boldsymbol{X}$, the proposed M2SA module is less computationally expensive than the traditional multihead self-attention module. In addition, $\boldsymbol{K}$ and $\boldsymbol{V}$ contain rich multiscale information, and the proposed M2SA module is more capable of modeling global contextual dependencies, which facilitates semantic segmentation. Note: At the end, we transform the resulting self-attention feature into a 2-D feature map as output.

Inspired by SENet [45], we design a channel attention branch to efficiently compute channel attention. Unlike the SE module, we replace the FC layer with a 1×1 convolutional layer to better fit the 2-D image structure. For the second branch, the input feature $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$ is first fed into a global average pooling layer to obtain the channel attention mapping $\boldsymbol{X}_p \in \mathbb{R}^{C\times1\times1}$. After a 1×1 convolution to reduce the channel dimension, passed through a ReLu6 activation function and a 1×1 convolution to recover the original channel dimension. Then a Sigmoid activation function is used to obtain the channel weight distribution, which is element-wise multiplied with the original input $\boldsymbol{X}$ to obtain the feature map incorporating the channel attention. In summary, the above process can be expressed as follows:

$$\begin{aligned} \boldsymbol{X}_p &= \text{Avgpool}(\boldsymbol{X}) \\ \boldsymbol{X}_c &= \text{ReLu6}\left(\text{Conv}_{1\times1}\left(\boldsymbol{X}_p\right)\right) \\ \text{Attention}_c &= \text{Sigmoid}(\text{Conv}_{1\times1}(\boldsymbol{X}_c)) \otimes \boldsymbol{X}. \end{aligned} \tag{8}$$

Finally, the features of the two branches are fused by a summation operation to obtain the final output features

$$\boldsymbol{X}_{\text{out}} = \text{Attention} + \text{Attention}_c. \tag{9}$$

C. E-FFN Module

FFN is an important component of the transformer for feature enhancement [44]. As shown in Fig. 4(a), traditional transformers typically rely on the FC layer as the FFN [22], [44] and rely entirely on attention to capture inter-pixel

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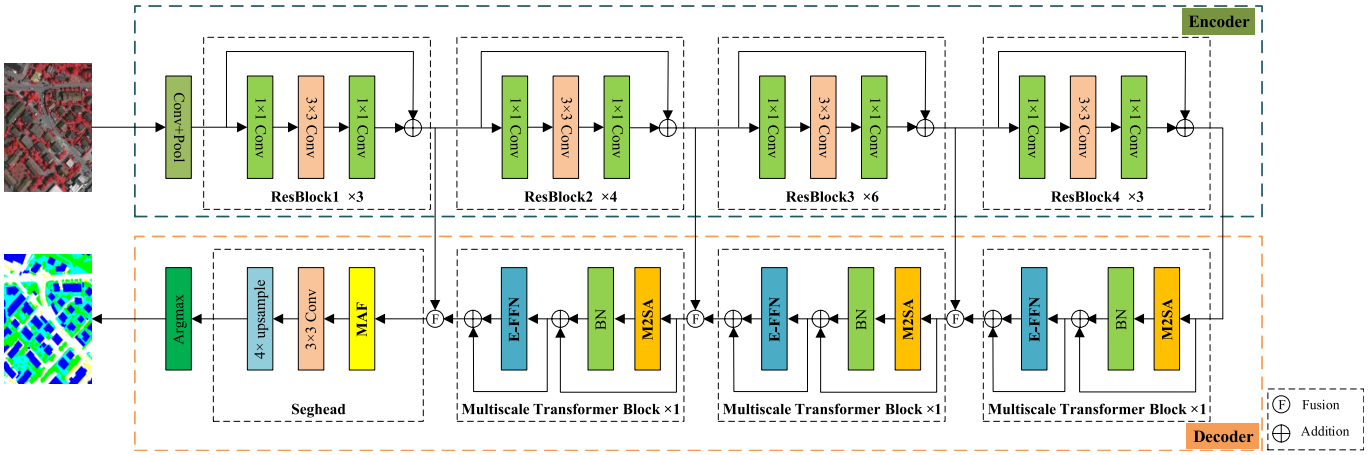

Fig. 2. Overall network structure of the CMTFNet.

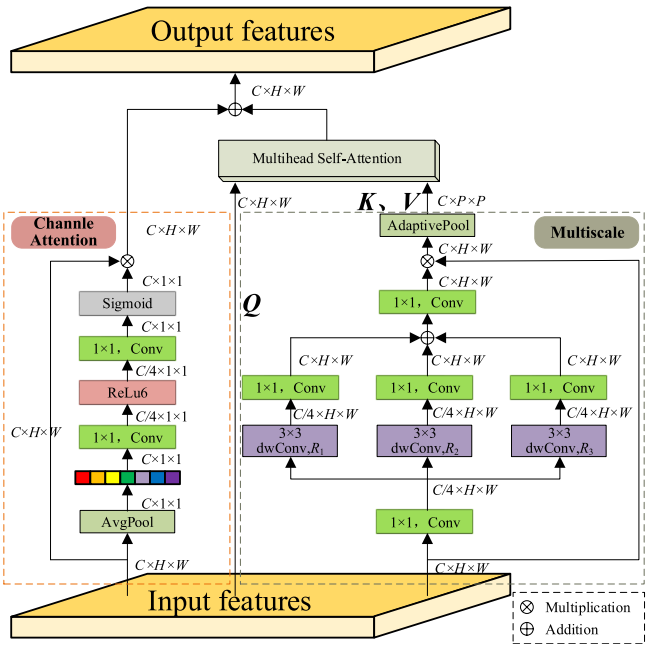

Fig. 3. Illustration of the M2SA module.

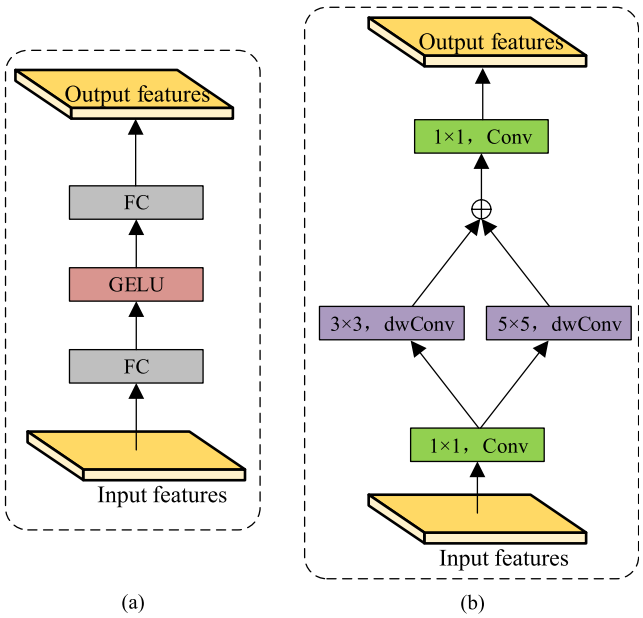

Fig. 4. (a) Comparison of the original FFN. (b) Our E-FFN.

dependencies. Though effective, this design is not good at learning the local information of 2-D images, which is also important for semantic segmentation. Therefore, we replace the FC layer in the FFN with a 1×1 convolution and insert two parallel 3×3 and 5×5 depthwise separable convolutions in between, as shown in Fig. 4(b). So the FFN network can inherit the local extraction capability of the CNN. In summary, the process can be formulated as follows:

$$\begin{aligned} \boldsymbol{X}_1 &= \mathrm{dwConv}_{3\times3}(\mathrm{Conv}_{1\times1}(\boldsymbol{X})) \\ \boldsymbol{X}_2 &= \mathrm{dwConv}_{5\times5}(\mathrm{Conv}_{1\times1}(\boldsymbol{X})) \\ \boldsymbol{X}_{\mathrm{out}} &= \mathrm{Conv}_{1\times1}(\boldsymbol{X}_1 + \boldsymbol{X}_2) \end{aligned} \tag{10}$$

where $\boldsymbol{X}$ denotes the input feature, $\boldsymbol{X}_i$ $(i = \{1, 2\})$ denotes the output of two depthwise separable convolutions, $\mathrm{Conv}_{1\times1}(\cdot)$ denotes a 1×1 convolution, and $\mathrm{dwConv}_{n\times n}(\cdot)$ denotes an $n \times n$ depthwise separable convolution. The multiscale transformer block is constructed by the M2SA module and the E-FFN module defined above.

D. MAF Module

The shallow features generated by ResBlock1 are rich in spatial detail information, such as color, texture, and edges, but lack semantic information. The last multiscale transformer block generates deep features that are rich in semantic information but have a coarse spatial resolution. Therefore, to effectively fuse these two features, we propose the MAF module, as shown in Fig. 5.

We first perform a weighted fusion of the two features to take full advantage of the precise spatial detail information and semantic information. The fused features are then used as input to the MAF. To further improve segmentation accuracy, we apply three branches to the input feature $\boldsymbol{F} \in \mathbb{R}^{C\times H\times W}$ to enhance the representation of channel, multiscale, and spatial features, where C denotes the channel dimension, and H and W denote the feature map size.

Specifically, for the channel branch, same as the channel attention branch in the M2SA module, a global average pooling is first used to generate the channel attention map

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$\boldsymbol{F}_p \in \mathbb{R}^{C\times1\times1}$. The channel dimension is reduced by a 1×1 convolution, restored to the original channel dimension after a ReLu6 activation function and a 1×1 convolution. Then, a Sigmoid activation function is used to obtain the channel weight distribution, which is element-wise multiplied with the original input to obtain the feature map incorporating the channel attention. The multiscale branch first uses a 1×1 convolution to reduce the channel dimension and extracts multiscale information through three parallel depthwise separable atrous convolutions, where the convolution kernel size is 3×3, and the atrous rate R_i ($i = \{1, 2, 3\}$) is $\{7, 9, 11\}$. After being restored to the original channel dimension by a 1×1 convolution, the three parallel results are subsequently fused by a summation operation. The output of the multiscale branch is obtained by a 1×1 convolution and residual connection. Similar to the spatial attention module in the convolutional block attention module (CBAM) [46], we aggregate the information in the channel dimension to generate an efficient feature descriptor. The spatial branch first generates the feature map using max pooling and average pooling and then concatenates in the channel dimension to generate a spatial attention map $\boldsymbol{F}_s \in \mathbb{R}^{2\times H\times W}$. The spatial weight distribution is obtained by a 5×5 convolution and a Sigmoid activation function, which is element-wise multiplied with the original input to obtain the feature map containing spatial attention. Finally, the results of the three branches are fused by a summation operation to produce the final feature map.

IV. Dataset and Experimental Setting

A. Datasets

1) ISPRS Vaihingen: This dataset contains 33 orthophotos, which have an average size of 2494×2064 and a ground sampling distance (GSD) of 9 cm. Each image contains three bands, near-infrared, red, and green, along with the corresponding digital surface model (DSM) and normalized DSM (NDSM). It consists of six categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background.

2) ISPRS Potsdam: This dataset contains 38 orthophotos with an image size of 6000×6000 and a GSD of 5 cm. Each image contains near-infrared, red, green, and blue bands, as well as the corresponding DSM and NDSM. As with the Vaihingen dataset, six categories are included: impervious surfaces, buildings, low vegetation, trees, car, and clutter/background.

B. Evaluation Metrics

To evaluate our proposed model, we used three evaluation metrics, including overall accuracy (OA), the mean intersection over union (mIoU), and mean F1 score (mF1), to compare with currently popular methods. Based on the accumulated confusion matrix, OA, mIoU, and mF1 are calculated as follows:

$$\mathrm{OA} = \frac{\sum_{k=1}^{N} \mathrm{TP}_k}{\sum_{k=1}^{N} \mathrm{TP}_k + \mathrm{FP}_k + \mathrm{TN}_k + \mathrm{FN}_k} \tag{11}$$

$$\mathrm{mIoU} = \frac{1}{N}\sum_{k=1}^{N} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k} \tag{12}$$

$$\mathrm{precision} = \frac{1}{N}\sum_{k=1}^{N} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k} \tag{13}$$

$$\mathrm{recall} = \frac{1}{N}\sum_{k=1}^{N} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k} \tag{14}$$

$$\mathrm{F1} = 2 \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \tag{15}$$

where TP_k, FP_k, TN_k, and FN_k denote true positives, false positives, true negatives, and false negatives, respectively, for object indexed as class k. OA is the ratio of the number of correctly predicted pixels to the total number of pixels. Also, we used the number of floating-point operations per second (Flops) and the number of params (for processing $3 \times 512 \times 512$ images) to evaluate the complexities of the model.

C. Experimental Setting

Considering that many remote-sensing datasets do not have DSM and NDSM, we did not use DSM and NDSM in all experiments. We divided the dataset according to the official advice given, for the ISPRS Vaihingen dataset, we used 16 images for training and 17 images for testing. For the ISPRS Potsdam dataset, 24 images were used for training and 14 images were used for testing. To prevent over-fitting, we cropped the training data to 512×512 size with a sliding window operation in steps of 256, and then performed data augmentation using random horizontal flip, random vertical flip, and random multiscale [0.5×, 0.75×, 1×, 1.25×, 1.5×] crop.

All experiments were trained and validated on a single NVIDIA Tesla V100S GPU using the PyTorch framework to ensure a fair comparison. In the training phase, we used the AdamW optimizer for training. The weight decay parameter is 0.01, the initial learning rate is 1e-4, the loss function is cross-entropy, and the learning rate is updated using a "poly" learning strategy [15], where the power is 0.9. The training batch size is 8, and the max training count is 200. In the testing phase, a multiscale [0.5×, 0.75×, 1×, 1.25×, 1.5×] test time augmentation (TTA) strategy was used.

V. Experimental Results and Analysis

A. Comparison With State-of-the-Art Methods on ISPRS Vaihingen

The experimental results of the different methods on the ISPRS Vaihingen dataset are shown in Table I. As with the other experiments, the accuracy of clutter/background is not reported. This is because the Vaihingen dataset has less clutter/background. The results in Table I show that our proposed CMTFNet method obtained the highest MeanF1/mIoU/OA. Compared with the suboptimal method, MeanF1 improved by 0.55%, mIoU by 0.82%, and OA by 0.35%. And our method obtained the highest scores in all categories. In particular, on the small-scale target car category,

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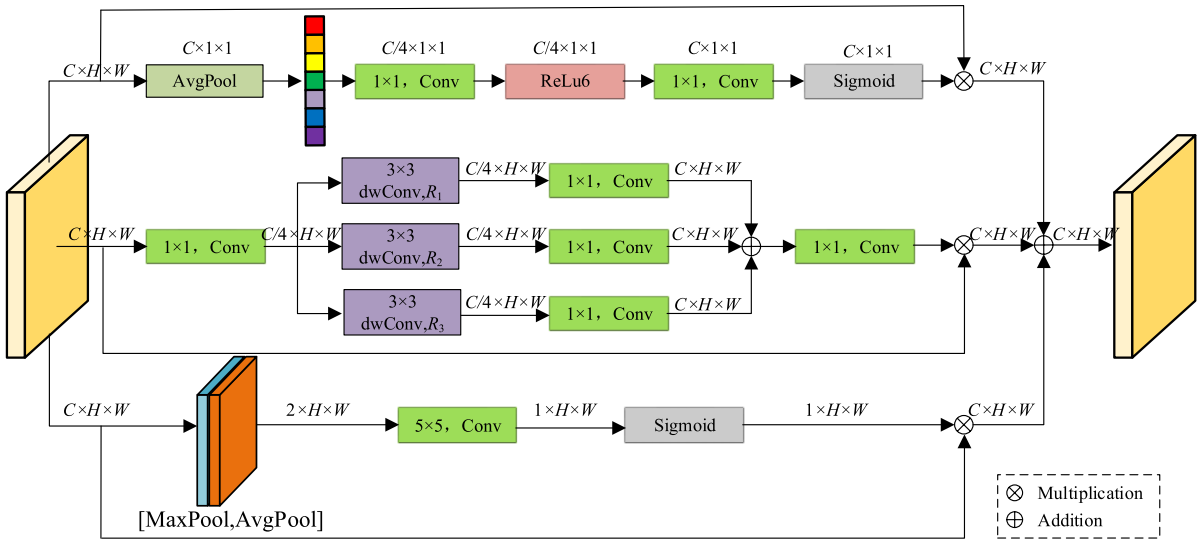

Fig. 5. Illustration of MAF module.

our method obtained an F1 score of 82.77%, outperforming the second-placed Swin-UperNet by 1.19%.

To visualize the differences between our method and other popular methods, we show the visualization results for each method in Fig. 6. As can be seen in Fig. 6, CMTFNet achieves better segmentation results than other methods. When faced with the large-scale target "building category," CMTFNet can segment the complete outline of the target object very well. In the case of the small-scale target "car category," CMTFNet can segment each car more accurately than the other methods. It can be seen that CMTFNet can achieve good segmentation results for different scales of targets.

B. Comparison With State-of-the-Art Methods on ISPRS Potsdam

We have compared this with currently popular methods on a larger dataset. The experimental results of the different methods on the ISPRS Potsdam dataset are shown in Table II. As with the Vaihingen dataset, the accuracy of the clutter/background is not reported. As can be seen from the results in Table II, our method obtained the highest MeanF1, mIoU, and OA. Compared to the suboptimal method, MeanF1 improved by 0.32%, mIoU by 0.52%, and OA by 0.43%. And our method achieved the highest scores in all categories. It even achieved an F1 score of 96.41% in the building category and an F1 score of 92.41 in the car category.

Also, to visualize the differences between our method and other popular methods, we show the visualization results for each method in Fig. 7. As can be seen from Fig. 7, our proposed CMTFNet captures more detailed information and can accurately identify the edges of objects compared to other methods.

C. Ablation Experiments

1) Effect of Each Module of CMTFNet: To evaluate the performance of each module of the proposed CMTFNet, ablation experiments were carried out on the ISPRS Vaihingen dataset. The effectiveness of each module was tested by overlaying it in turn. The experimental results are shown in Table III. Baseline: The baseline was constructed by the FCN8S with ResNet50 as the backbone network.

a) Baseline + M2SA: Three multiscale transformer blocks act as decoders for baseline, but we used the FFN of the traditional transformer as the FFN of the multiscale transformer block. As shown in Table III, by adding the M2SA module, performance improved for all categories, with the network's MeanF1 improving by 1.36%, mIoU by 1.91%, and OA by 0.38%. In particular, for the car category, the F1 score improved by 5.71%, which demonstrates the effectiveness of the M2SA module.

b) Baseline + M2SA + E-FFN: We used the complete multiscale transformer block as the decoder. As can be seen from Table III, the mIoU of the network improved by 0.38% after the introduction of the E-FFN module, which proves the effectiveness of the E-FFN module.

c) CMTFNet: The MAF module was introduced to build the full CMTFNet network. As can be seen from Table III, the CMTFNet network achieved the best results for most categories. In particular, for the car category, an F1 score of 82.77% was achieved, an improvement of 0.94%, and obtained the highest MeanF1, mIoU, and OA.

We also visualize the results for each module, as shown in Fig. 8. When the multiscale transformer block and MAF modules were not added, the network lacked the ability to model global contextual information, resulting in the inability to accurately identify car categories. When all modules were added, the network was able to effectively extract global contextual information and multiscale information, enhancing the network's ability to understand multiscale scenes and improving its ability to recognize the edges of objects.

2) Effect of M2SA Module: This part explores the performance of the multiscale branch and channel attention branch in the M2SA module. The experimental results are shown in Table IV. When only the multiscale branch was added, the mIoU was reduced by 0.54% due to the lack

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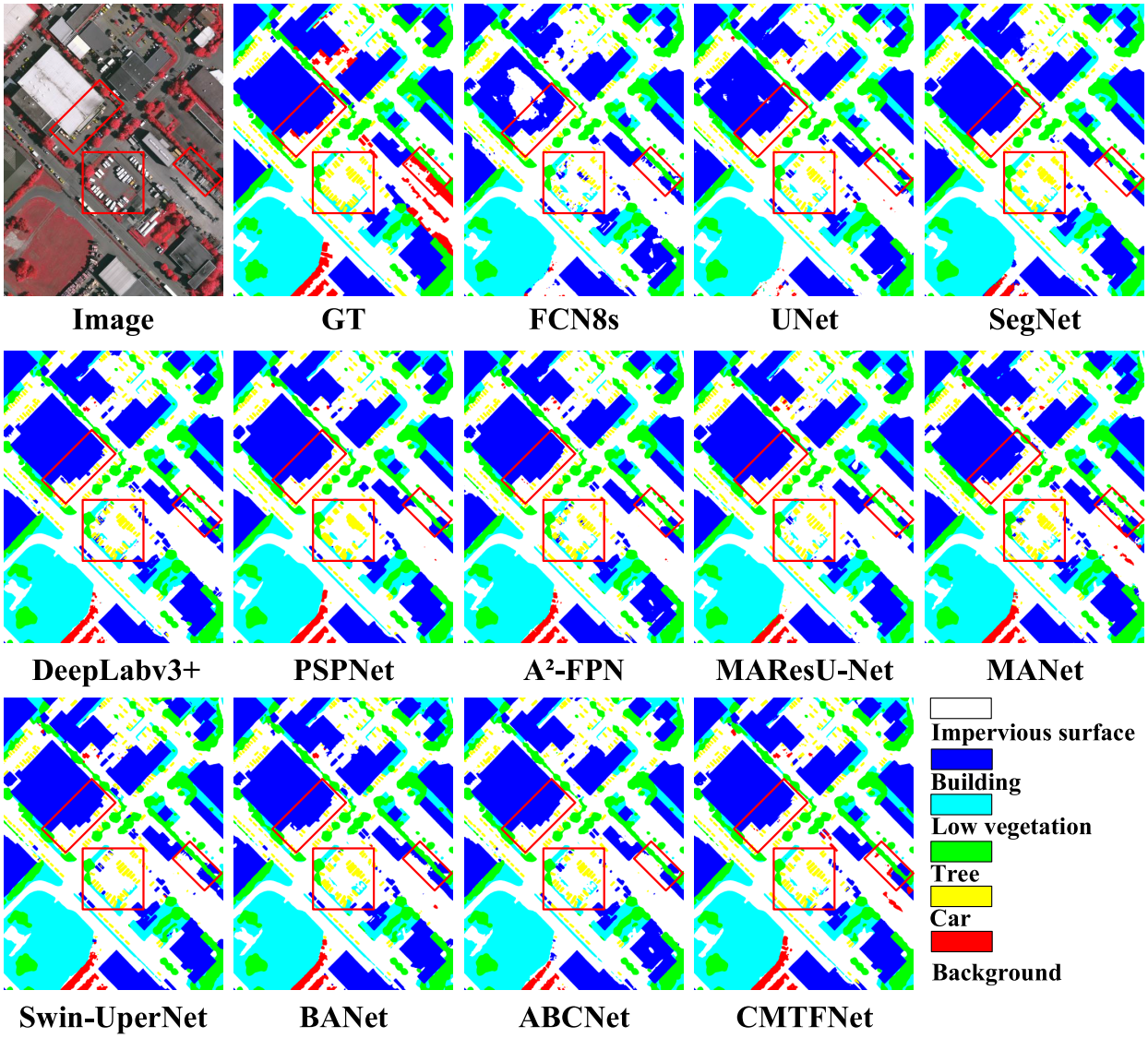

Fig. 6. Qualitative comparison of the visualization results of our method with other methods on the Vaihingen dataset.

TABLE I

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE ISPRS VAIHINGEN DATASET. THE BEST VALUES IN THE COLUMNS ARE IN BOLD. ALL SCORES ARE EXPRESSED AS PERCENTAGES (%). WHERE THE METRIC FOR ALL CATEGORIES IS F1 SCORES

Method	Imp.surf.	Building	Lowveg.	Tree	Car	MeanF1	mIoU	OA
FCN-8S [12]	87.94	90.74	78.45	85.83	73.04	83.20	71.75	85.71
UNet [13]	89.86	93.41	80.82	86.93	81.50	86.51	76.55	87.79
SegNet [31]	89.35	93.14	80.57	86.75	74.37	84.84	74.23	87.41
DeepLabv3+ [17]	90.40	94.06	81.45	87.20	81.23	86.87	77.13	88.36
PSPNet [18]	89.99	93.78	81.51	87.41	79.28	86.39	76.44	88.23
A²-FPN [33]	90.09	93.69	81.03	87.08	80.14	86.41	76.44	88.05
MAResU-Net [36]	90.43	94.13	81.73	87.17	80.18	86.73	76.95	88.39
MANet [37]	89.99	93.53	81.25	87.37	80.83	86.59	76.69	88.12
Swin-UperNet [26]	90.26	94.07	81.13	87.04	81.58	86.82	77.05	88.21
BANet [42]	90.28	93.86	80.99	87.08	80.31	86.50	76.60	88.12
ABCNet [47]	89.73	93.38	80.91	87.23	78.72	85.99	75.83	87.86
CMTFNet (ours)	**90.61**	**94.21**	**81.93**	**87.56**	**82.77**	**87.42**	**77.95**	**88.71**

of channel information interaction. The params are reduced by 0.39 M and the reduction of FLOPs is negligible. It demonstrates the importance of the channel attention branch in the M2SA module and does not introduce significant

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TABLE II

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE ISPRS POTSDAM DATASET. THE BEST VALUES IN THE COLUMNS ARE IN BOLD. ALL SCORES ARE EXPRESSED AS PERCENTAGES (%). WHERE THE METRIC FOR ALL CATEGORIES IS F1 SCORES

Method	Imp.surf.	Building	Lowveg.	Tree	Car	MeanF1	mIoU	OA
FCN-8S [12]	90.24	94.70	84.55	85.90	90.55	89.19	80.68	87.91
UNet [13]	90.73	95.35	85.05	85.97	91.49	89.72	81.57	88.36
SegNet [31]	91.27	95.18	85.10	86.05	91.10	89.74	81.60	88.54
DeepLabv3+ [17]	91.76	96.33	85.74	86.87	92.23	90.59	83.02	89.45
PSPNet [18]	91.74	96.32	85.80	86.97	91.86	90.54	82.94	89.42
A²-FPN [33]	91.82	96.11	85.65	86.86	92.11	90.51	82.89	89.40
MAResU-Net [36]	91.79	96.33	85.69	87.03	92.19	90.61	83.05	89.46
MANet [37]	91.71	96.09	85.72	87.26	92.16	90.59	83.01	89.42
Swin-UperNet [26]	91.60	96.04	86.09	87.00	91.70	90.49	82.82	89.43
BANet [42]	91.42	95.65	85.67	86.88	91.40	90.20	82.35	89.14
ABCNet [47]	91.21	95.92	85.28	86.56	90.74	89.94	81.94	88.94
CMTFNet (ours)	**92.12**	**96.41**	**86.43**	**87.26**	**92.41**	**90.93**	**83.57**	**89.89**

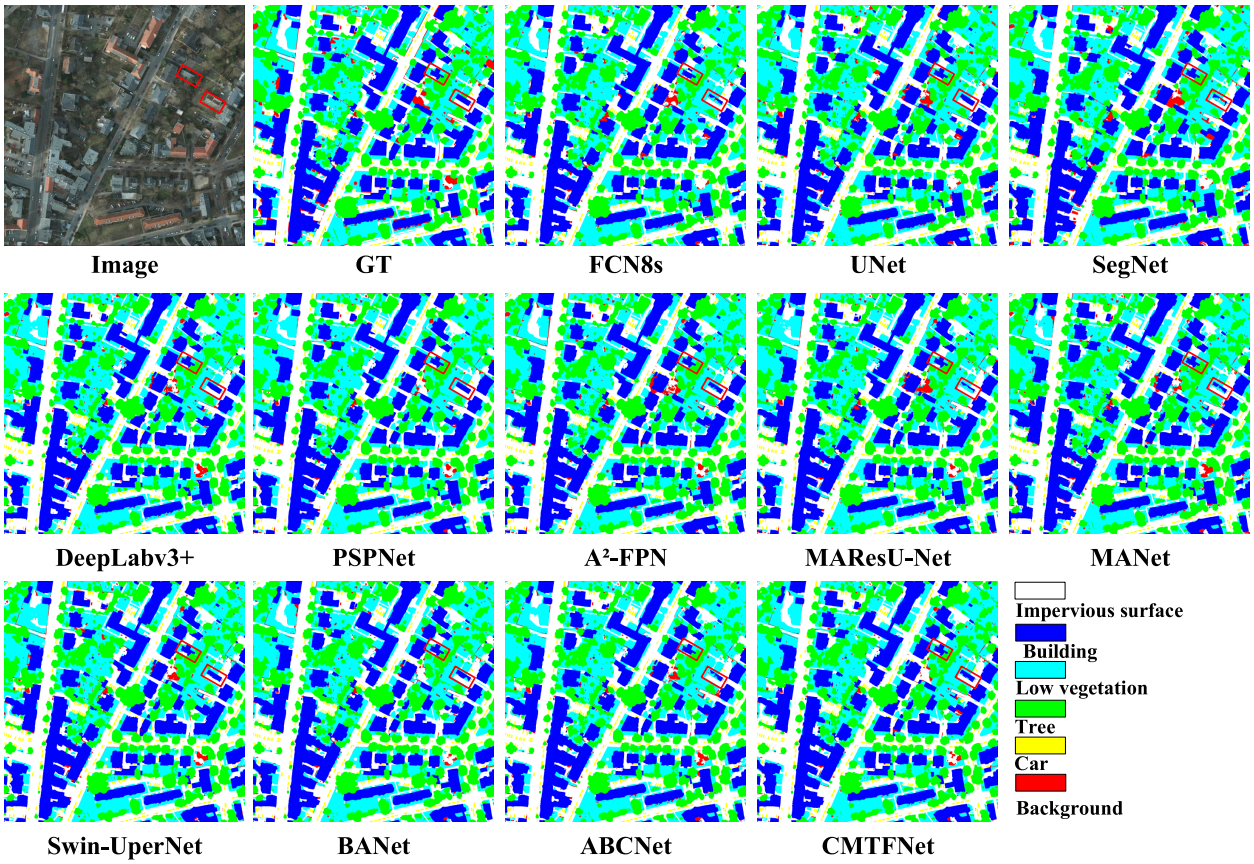

Fig. 7. Qualitative comparison of the visualization results of our method with other methods on the Potsdam dataset.

TABLE III

ABLATION STUDY OF EACH COMPONENT OF THE CMTFNET. THE BEST VALUES IN THE COLUMNS ARE IN BOLD. ALL SCORES ARE EXPRESSED AS PERCENTAGES (%), WHERE THE METRIC FOR ALL CATEGORIES IS F1 SCORES

Method	Imp.surf.	Building	Lowveg.	Tree	Car	MeanF1	mIoU	OA
Baseline	90.11	93.55	81.42	87.48	75.56	85.62	75.40	88.11
Baseline+M2SA	90.35	94.20	81.56	87.51	81.27	86.98	77.31	88.49
Baseline+M2SA+E-FFN	**90.63**	94.17	**82.01**	87.53	81.83	87.23	77.69	88.67
CMTFNet	90.61	**94.21**	81.93	**87.56**	**82.77**	**87.42**	**77.95**	**88.71**

computational overhead. When only the channel attention branch was added, the mIoU was reduced by 0.78% due to the lack of multiscale semantic information. The params are reduced by 1.61 M and the FLOPs are reduced by 2.87 G. This

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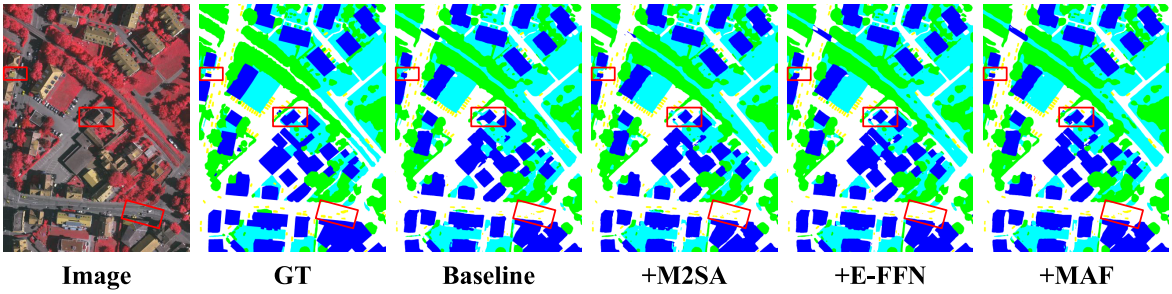

Fig. 8. Visualization of ablation experiments on the Vaihingen dataset.

TABLE IV
RESULTS OF ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET FOR THE M2SA MODULE

Method	Multiscale	Channel	Params (Mb)	FLOPs (Gbps)	mIoU
M2SA	✓		29.68	32.85	77.41
M2SA		✓	28.46	29.98	77.17
M2SA	✓	✓	30.07	32.85	77.95

TABLE V
RESULTS OF ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET FOR THE E-FFN MODULE

Method	3 × 3	5 × 5	Params (Mb)	FLOPs (Gbps)	mIoU
FFN			30.05	32.82	77.53
E-FFN	✓		30.06	32.84	77.57
E-FFN		✓	30.07	32.85	77.65
E-FFN	✓	✓	30.07	32.85	77.95

TABLE VI
RESULTS OF ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET FOR THE MAF MODULE. (W/O DENOTES WITHOUT)

Method	Params (Mb)	FLOPs (Gbps)	mIoU
w/o Channel	30.04	32.85	77.74
w/o Multiscale	29.93	30.62	77.69
w/o Spatial	30.07	32.85	77.84
MAF	30.07	32.85	77.95

shows the importance of multiscale information in semantic segmentation.

3) Effect of the E-FFN Module: This part explores the performance of convolution in the E-FFN module. The experimental results are shown in Table V. When only a 3 × 3 depthwise separable convolutions was used, the mIoU is reduced by 0.38%. When only a 5 × 5 depthwise separable convolutions was used, the mIoU is reduced by 0.3%. When both were used, the highest segmentation accuracy was obtained without additional computational overhead. In addition, we compared the E-FFN module with the traditional feedforward network FFN, and the experimental results demonstrate the effectiveness of the E-FFN module.

4) Effect of MAF Module: We performed ablation experiments on the three branches in the MAF module. The experimental results are shown in Table VI. The mIoU is reduced by 0.21% when the channel attention branch was removed, by 0.26% when the multiscale branch was removed, and by 0.11% when the spatial attention branch was removed. In addition, the computational overhead from the three branches is almost negligible. The experimental results further demonstrate the effectiveness of the MAF module.

TABLE VII
COMPARISON OF THE COMPLEXITIES OF OUR METHOD WITH OTHER METHODS. THE BEST VALUES IN THE COLUMNS ARE IN BOLD

Method	Params (Mb)	FLOPs (Gbps)	mIoU
FCN-8S [12]	33.63	85.10	71.75
UNet [13]	24.15	219.62	76.55
SegNet [31]	29.48	563.17	74.23
DeepLabv3+ [17]	40.35	138.63	77.13
PSPNet [18]	53.32	28.00	76.44
A²-FPN [33]	22.82	41.73	76.44
MAResU-Net [36]	26.28	35.09	76.95
MANet [37]	35.86	77.62	76.69
Swin-UperNet [26]	58.89	234.86	77.05
BANet [42]	**12.72**	**13.00**	76.60
ABCNet [47]	13.39	15.58	75.83
CMTFNet (ours)	30.07	32.85	**77.95**

D. Model Complexities

Table VII shows the computational complexities and parameter sizes of all methods under the same operating environment. From Table VII, it can be seen that CMTFNet achieves superior performance with low computational complexity compared to the other methods.

VI. CONCLUSION

In this article, we present CMTFNet, a hybrid model with CNN and a multiscale transformer for remote-sensing image semantic segmentation. CMTFNet combines the advantages of CNN and transformer to learn local features and global features and facilitates the learning of multiscale feature representations through the MSA module M2SA embedded in the transformer. Moreover, we propose an MAF module at the end of the decoder to achieve efficient coupling of deep and shallow features. We have demonstrated the effectiveness of our proposed method by conducting extensive comparative tests and ablation experiments on two high-resolution remote-sensing datasets—ISPRS Vaihingen and ISPRS Potsdam.

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In future research, we will further optimize our network structure and will continue to explore the potential and applications of CNN and transformers in the field of remote sensing.

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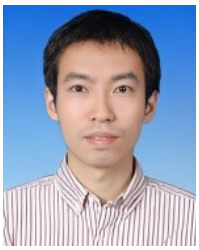

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